

1. Visual Design Logic

The visuals were selected based on one goal: **making patterns obvious without wasting screen real estate.**

- **Line Charts (Login Frequency vs. Session Duration / Time Trends):**
Used because both metrics are continuous and the relationships matter only when seen across ranges or time. The line makes growth, drops, and anomalies instantly visible.
- **Bar/Column Charts (Counts by Device Type, Customer Segments):**
Best for categorical comparison. You need quick ranking — bars do that without cognitive load.
- **Pie/Donut (If used):**
Only for proportional splits where the user needs a quick “share of total” view. Not for detailed comparison.
- **Cards / KPIs:**
Used to highlight essential numbers like total customers, average duration, or high-risk counts. These are values users must see instantly.

The visuals were chosen strictly based on **data type** and **analysis intent**, not aesthetics.

2. Key Insights

These come directly from the graphs and your metrics — not generic corporate nonsense.

Customer Behavior

- **Session duration increases with login frequency**, but not linearly.
There's a sharp rise at mid-range frequencies, then it stabilizes.
→ Heavy users behave differently from casual users.
- Median session duration shows **significant jumps** at certain frequency levels, hinting at **clustered behavior groups** (low-, mid-, and high-engagement users).

Device-Type Breakdown

- One or two device types dominate customer count.

Risk Segmentation

- After cleaning the anomaly column, high-risk customers form a smaller but important segment.
Their distributions across device type or login habit can flag vulnerabilities.

Engagement Trends

- Spikes in session duration usually align with increased login count rather than device differences.
- No consistent decline pattern — meaning engagement is not deteriorating but fluctuating.

3. Performance Metrics (DAX-Driven Insights)

- **High_Risk Anomaly Count:**
Identifies how many customers fall into high-risk behavior. Sudden increases indicate potential fraud clusters.
- **Average Transaction Amount:**
Shows spending patterns across segments. Sharp jumps usually signal abnormal or suspicious activity.
- **Customers per Device Type:**
Reveals customer distribution by device. Imbalances or sudden shifts may indicate bot activity or device-specific misuse.
- **Fraud Rate:**
Measures the proportion of fraud in total transactions. Rising trends highlight emerging vulnerabilities.
- **Total Amount:**
Tracks overall transaction volume. Useful for spotting revenue peaks or unexpected drops in activity.
- **Total Fraud Transactions:**
Shows the absolute scale of fraudulent actions. Spikes pinpoint high-risk time periods needing investigation.

These DAX-based metrics expose stability issues, user-type clusters, and platform performance differences.

4. Recommendations

This is where you convert insights into action — not generic “improve performance” garbage.

a. Target High-Engagement Users

Your graph shows mid-range login users suddenly jump in session duration.
→ Focus feature upgrades, promotions, and retention features around this group.
They’re giving the strongest return.

b. Fix Underperforming Device Segments

If one device type is far behind:

- Test UI flow and load time.
- Fix compatibility issues.
- Optimize UX for that device class.

c. Risk-Based Monitoring

Since moderate risk rows were removed and high/low are clean:

- Track high-risk users’ behavior by device type and login pattern.

- Identify whether certain devices correlate with more anomalies.

d. Improve Onboarding for Low-Frequency Users

Low-frequency users have short session times → they're not finding value.
Redesign onboarding, tutorials, or home screen shortcuts for them.

e. Build Predictive Alerts

Your engagement patterns can feed:

- churn prediction,
- anomaly detection,
- user retention forecasting.

Small investment, large payoff.